

“Measurement” of Tinnitus

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Chronic tinnitus is the persistent sensation of hearing a sound that exists only inside the head. The prevalence of tinnitus in adults in the United States is estimated at 10 to 15%. For about 20% of these individuals the tinnitus is significantly bothersome. Although myriad therapies for tinnitus are offered (often at significant cost), most are not evidence based. Difficulty in the assessment and further development of interventions for tinnitus stems from the limitations of techniques used to evaluate these interventions. Questionnaires are widely available to “measure” (tinnitus can only be indirectly measured) functional effects of tinnitus, such as difficulty sleeping and concentrating, and negative emotions such as anxiety, depression, and annoyance. Questionnaires have recently been documented for sensitivity to change in response to intervention (i.e., “responsiveness”). All of these questionnaires function well to assess the overall impact of tinnitus. The

limitations mentioned pertain primarily to measures of tinnitus perception, which typically include the psychoacoustic measures of tinnitus loudness and pitch matches, tinnitus spectral content, minimum masking levels, and residual inhibition. These measures, which are obtained routinely in many clinics and as part of research studies, have not been validated for being diagnostic, prognostic, discriminative, or responsive. In order for these measures to become clinically meaningful, normative standards are needed, both for baseline measures and for repeated measures of tinnitus perception. Evidence-based intervention for tinnitus requires accurately measuring both the perception of, and reactions to, tinnitus.
Key Words: Diagnosis—Measurement—Questionnaire—Tinnitus.

Otol Neurotol 37:e276–e285, 2016.

Tinnitus is the persistent sensation of sound for which no acoustic source for the sound exists outside of the head. When a patient complains of tinnitus, it is essential to determine if tinnitus-specific clinical services are required. Tinnitus assessments are most typically performed by otolaryngologists, audiologists, and mental health providers. Regardless of who performs the assessment, it is also essential to determine if a referral is needed for further medical services, and the type and extent of intervention that may be required to mitigate any negative reactions associated with the tinnitus (1).

The identification of tinnitus currently relies on patient report—there is no objective means of identifying the presence of tinnitus. Identifying the symptom, however, does not address whether or not a person’s tinnitus is problematic (2). Whereas tinnitus is experienced by 10 to 15% of all adults, only about one in five of those who experience tinnitus seek professional services (3–5). For the remainder, the symptom is essentially innocuous.

These people pay little to no attention to their tinnitus, and their everyday life is unaffected by the phantom auditory symptom.

It is well accepted that tinnitus is usually associated with some degree of reduced auditory sensitivity (6,7). Because of the common nexus between tinnitus and hearing loss, it is important that anyone experiencing tinnitus undergo a hearing assessment by an audiologist (8). An audiologist would know if a medical referral was indicated. The audiologist should also be capable of determining if intervention is needed specifically for the tinnitus.

Interventions for tinnitus target: 1) the perception of tinnitus and/or 2) reactions to tinnitus. Perception of tinnitus refers to the auditory perceptual characteristics of tinnitus (loudness, pitch, timbre/spectral content). Interventions targeting tinnitus perception generally attempt to reduce the loudness (or intensity) of tinnitus. Tinnitus researchers and clinicians often obtain measures of tinnitus perception, including psychoacoustic measures and various rating scales (9). Reactions to tinnitus refer to the various domains of tinnitus impact, such as emotional distress, concentration difficulties, reduced sense of control, sleep disturbance, and others. Numerous questionnaires have been developed and validated for assessing the negative impact of tinnitus.

The purpose of this article is to review techniques that exist for measuring both perception of, and reactions to tinnitus. Because tinnitus perception and reactions are

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Dr. J.A.H. is supported by a Research Career Scientist award provided by VA Rehabilitation Research & Development (RR&D) Service (C9247S).

The author discloses no conflicts of interest.

DOI: 10.1097/MAO.0000000000001070

subjective phenomena, use of the term “measure” indicates any attempt to quantify subjective sensations or reactions that are associated with tinnitus. Measurement of tinnitus is an indirect process—direct, or objective, quantification of tinnitus parameters is currently not possible.

TYPES OF TINNITUS

The definition of tinnitus may seem obvious, but there are nuances. Almost everyone experiences transient ear noise—a tonal- or whistle-type sound that emerges suddenly and seemingly at random in one ear, and then decays. The sound is accompanied by a sense of ear fullness and hearing loss. After about a minute all symptoms resolve. Transient ear noise is a normal occurrence and should not be considered tinnitus (1).

At the most fundamental level, there are two types of tinnitus. By far the most common type is generated within the auditory pathways and consists entirely of neural activity. This has been referred to as *neurophysiologic* tinnitus, which is most commonly caused by exposure to loud noise, but can also be caused by ototoxicity, head trauma, whiplash injury, and numerous medical conditions (2). Neurophysiologic tinnitus is also referred to as *sensorineural* or just *neural* tinnitus. In the clinical practice guideline recently published by the American Academy of Otolaryngology—Head and Neck Surgery Foundation, this type is referred to as *primary* tinnitus (8).

Much less common is tinnitus that is generated as an acoustic signal somewhere in the head or neck, which is referred to as *somatic* tinnitus (or *secondary* tinnitus in the American Academy of Otolaryngology—Head and Neck Surgery Foundation guideline) (8). The origin of somatosounds can be muscular, respiratory, skeletal, or vascular structures (10). The temporomandibular joint can also be the source of somatosounds (11). The most common somatosound is pulsatile tinnitus, typically a “whooshing” sound that pulses in synchrony with the heartbeat (12). In most patients of pulsatile tinnitus, the condition is benign. However, serious conditions are possible and should be considered. A patient with suspected pulsatile tinnitus, or any form of nonpulsatile somatic tinnitus, should be referred for a medical examination by an otolaryngologist (13).

Subjective tinnitus is heard only by the person with tinnitus. Objective tinnitus, which is very rare, means an examiner can also hear the person’s tinnitus (14). Tinnitus that can be heard by an examiner would by necessity be somatic tinnitus, i.e., the presence of an actual sound emanating from the ear canal would mean an acoustic event is taking place to cause the sound. The term “objective tinnitus” has also been used to mean the same thing as somatic tinnitus (15).

A common “textbook” definition of tinnitus, which serves to distinguish it from transient ear noise, is an internal sound lasting at least 5 minutes and occurring at least twice a week (16). This definition would be irrelevant for the great majority of patients who experience

constant tinnitus, which is always present even if it is not noticed or is masked by environmental sound. If constant tinnitus is suspected, an appropriate question to ask is “can you usually hear your tinnitus if you listen for it in a quiet room?” It would be unusual for a patient who experiences constant tinnitus to be unable to hear the tinnitus when sitting in a quiet room.

Tinnitus becomes chronic when it has been present for at least 6 months (8). This definition implies that tinnitus of less than 6 months duration would be considered acute or recent-onset tinnitus. Chronic tinnitus is also referred to as *permanent* or *persistent* tinnitus. There is also temporary tinnitus, which can be associated with noise exposure or certain medications, that usually lasts for a few days or up to a week before it resolves (17). As for temporary threshold shift (commonly known as TTS), temporary tinnitus is a sign that damage has been done to the cochlea.

Tinnitus is also reported as *intermittent*—typically described as switching between being “on” (present) and “off” (absent). The timing/periodicity of intermittent tinnitus has not been studied, but would likely be highly variable both within and across individuals who report this type of tinnitus. Because of the many factors that can affect whether or not tinnitus is perceived at a given moment in time, it is a challenge to confirm that a person’s tinnitus is actually intermittent. People may claim that their tinnitus switches on and off when in fact the “intermittency” may only pertain to their awareness of the tinnitus, which is typically affected by the acoustic environment and by mental distraction. A person’s acoustic environment changes throughout the day. As the level (and spectrum) of ambient sound changes, the tinnitus percept can alternate between being altered, masked, or unmasked. The presence of sound in the environment can thus eliminate the tinnitus (if the tinnitus is masked), or make it less noticeable because of the passive listening effect. Also, distraction can make a person completely unaware of the presence of tinnitus, even if the tinnitus is perceived as “loud.” It would of course be impossible to know if the tinnitus was actually present during periods of distraction. Any reports of intermittent tinnitus must be interpreted in light of these concerns, which would also apply to reports of tinnitus that “fluctuates in loudness.”

MEASURING REACTIONS TO TINNITUS

About 20% of people who experience tinnitus report that the sound adversely affects daily living such that clinical intervention is warranted (3,4). Numerous clinical studies have reported positive outcomes of various interventions for tinnitus; however, because methodology to measure such outcomes is inconsistent, statistical evidence supporting the effectiveness of tinnitus interventions remains inconclusive (18,19).

Dobie (2) described a pyramid analogy to conceptualize how tinnitus affects people differently. People who have tinnitus but are not bothered by it would be placed in the large base of the pyramid (cf. Fig. 1 of (20)). Above

that group would be people whose tinnitus is “bothersome,” with successively decreasing numbers of those with “mild,” “moderate,” and “severe” tinnitus. In the tip of the pyramid would be the relatively few people whose tinnitus is “debilitating,” meaning their tinnitus prevents them from living a normal life. This pyramid conceptualization illustrates how the majority of people who experience tinnitus do not require clinical services that are specific to the tinnitus. Those requiring such services have widely differing needs. In general, people whose tinnitus is not bothersome, or is only mildly bothersome, want assurance that their tinnitus is not symptomatic of a brain tumor or some other serious medical condition. For tinnitus that is more problematic, months, or even years of clinical intervention may be required. Determining where a patient would be placed in the continuum from nonbothersome to debilitating tinnitus requires the appropriate instruments for measuring tinnitus reactions, and proper timing for administering those instruments.

As mentioned, people with tinnitus are highly likely to also have hearing loss (6,7,21). Numerous researchers have suggested that some people with bothersome tinnitus blame their tinnitus for causing their hearing difficulties (2,22,23). This premise was tested prospectively by Ratnayake et al. (24) who systematically studied the relationship between complaints of hearing loss and complaints of bothersome tinnitus. These authors concluded “In tinnitus subjects, the awareness of impaired hearing may in fact be due to an underlying hearing loss rather than their tinnitus. In these cases, the impairment of hearing may contribute significantly to the perceived distress caused by the tinnitus” (p. 159).

It is clear that responses to tinnitus questionnaires may be influenced by hearing problems, resulting in index scores that are artificially inflated (reflecting a tinnitus problem that is worse than it actually is). The 10-item Tinnitus and Hearing Survey (THS) is a screening tool that was designed specifically to address this concern (20). The THS includes three subscales: A (Tinnitus), B (Hearing), and C (Sound Tolerance). The A (Tinnitus) subscale contains four items that address common tinnitus problems that would not be confused with a hearing problem. The score derived from the A subscale thus indicates a self-perceived problem that is specific to tinnitus. Similarly, the B (Hearing) subscale contains four items that address common hearing problems that would not be confused with a tinnitus problem. The score derived from the B subscale thus indicates a self-perceived problem that is specific to hearing. The scores obtained from a patient on the A and B subscales provide a realistic assessment of the relative difficulties caused by tinnitus and hearing loss.

The THS has been shown to be a valid and reliable clinical tool to assist in determining if intervention specific to tinnitus is appropriate. The THS was designed to be used in conjunction with a standard audiologic evaluation (25). Combined results from the evaluation and the THS will, in most patients, provide sufficient information to know if the patient requires tinnitus-

specific intervention. Full instructions for using the THS for this purpose have been published (20).

It should be mentioned that the THS will indicate a tinnitus-specific problem for some patients who also have hearing loss warranting amplification. It is well documented that hearing aids can often provide secondary tinnitus relief in such patients (8,26). If such a patient is to be fitted with hearing aids, then the patient should be reassessed with the THS after a reasonable period of time wearing the new aids (at least 1 month) to determine if the amplification was sufficient in remediating the tinnitus reactions.

If it is determined that a patient would benefit from tinnitus-specific intervention, then the next step in the “tinnitus measurement” process is to administer a tinnitus questionnaire. Numerous tinnitus questionnaires exist that have been validated for intake assessment, i.e., for scaling the negative impact of tinnitus (9). A fairly recent instrument, the Tinnitus Functional Index (TFI), has also been validated for being sensitive to changes in tinnitus reactions resulting from intervention (this sensitivity to change is referred to as *responsiveness*) (27). The TFI has garnered considerable international attention and, at last report, was being translated into at least 14 languages (28). It is conceivable that the TFI will attain acceptance as a “standard” tinnitus questionnaire, which would enable outcomes to be directly compared across different clinics and research studies. Instructions for administering, scoring, and interpreting scores obtained with the TFI are available (28).

In addition to the TFI, a Visual Numeric (loudness rating) Scale (VNS) can be administered to patients (29). Participants should complete the VNS in a quiet examination room (not a sound booth) before any audiometric or psychoacoustic testing to ensure that the rating of tinnitus loudness is not affected by auditory stimulation. Careful instructions are given to participants to ensure that only a vertical line is drawn on the scale (as compared with a circle or shaded area). They are instructed: “On the scale below, please draw a vertical line to indicate the loudness of your tinnitus at this moment.”

Although the VNS purports to assess tinnitus loudness, it is known that responses to such a scale will tend to correlate with the degree to which a person is impacted by his/her tinnitus (30). Patients who are more bothered by tinnitus tend to provide a higher rating on the VNS. Those who are less bothered tend to provide lower ratings. Of course, it is not possible to objectively measure the loudness of a person’s tinnitus, so the actual loudness of a person’s tinnitus can never be known with certainty. For these reasons, the VNS should not be considered a measure of tinnitus loudness per se, and is included as a measure of tinnitus reactions.

MEASURING TINNITUS PERCEPTION

Measures of tinnitus perception are routinely obtained by tinnitus researchers and clinical audiologists (31–35).

Over 30 years ago the Novartis Foundation (formerly the Ciba Foundation) in London spearheaded efforts to promote a standardized protocol for clinical tinnitus assessment (36,37). A key recommendation was that tinnitus measures should be standardized, including a psychoacoustic assessment battery of tinnitus to include pitch matching, loudness matching, minimum masking levels, and residual inhibition. Specific procedures for these tests had evolved through clinical efforts before 1975 (reviewed in [37]). Further development took place at the Oregon Hearing Research Center (at Oregon Health and Science University), and the protocol was adopted and advocated by the Ciba Symposium on Tinnitus in 1981.

Psychoacoustic measures of tinnitus are obtained for essentially two reasons: 1) to define the auditory attributes of tinnitus (i.e., what does it sound like for the individual?); and 2) to define the potential effects of external sound on tinnitus (i.e., when sound is applied to the ears of a person with tinnitus, does the external sound affect the auditory attributes of the tinnitus, and, if so, how?). The auditory attributes of tinnitus include loudness, pitch, and timbre/spectral composition. Effects of sound on tinnitus include masking/suppression effects, residual inhibition, exacerbation, and potentially some form of alteration. Testing that is commonly done to determine how sound affects tinnitus includes minimum masking levels and residual inhibition. There is currently no means of detecting or measuring exacerbation or alteration of tinnitus (other than patient report).

No scale exists to describe the degree of impairment corresponding to measurements of acoustic correlates of tinnitus; that is, the clinical measures of tinnitus are primarily descriptive relative to external sounds and are not informative regarding the site or extent of the pathology involved. Although the clinical value of tinnitus psychoacoustic measures is equivocal, they currently are used most commonly as a counseling tool—patients may feel validated that their tinnitus percept can be “quantified” numerically (and plotted on their audiogram) (38).

Tinnitus Loudness Measures

Clinically, the most important attribute of tinnitus is its loudness. Reducing the loudness of tinnitus would provide therapeutic benefit. The two most common methods of assessing tinnitus loudness are loudness matching and loudness ratings. We discussed above how tinnitus loudness ratings tend to reflect tinnitus impact (reactions) more than actual loudness (perception). We will focus here on tinnitus loudness matching, which involves presentation of an auditory stimulus (usually a pure tone) to the ear(s) and asking the individual to report if the stimulus is “louder or softer than the tinnitus.” The examiner adjusts the level of the stimulus until the tinnitus and the tone are matched in loudness.

Loudness matching originated with Fowler (39) who developed a test to balance the loudness of tinnitus in one ear with the loudness of a tone in the contralateral ear.

The level of the comparison tone, expressed in dB SL (Sensation Level, i.e., dB above the hearing threshold at the same frequency), was a measure of tinnitus loudness as experienced by the patient. Fowler noted that tinnitus loudness matches (LMs) were “paradoxically small,” i.e., they tended to be between about 5 and 10 dB SL (at the tinnitus pitch-matched frequency). He suggested there was a mismatch between the loudness as described by the patient and the more realistic loudness of the tinnitus as measured by loudness matching. Based on these results, he considered the loudness of tinnitus to be “faint.” Results similar to Fowler’s have been noted in virtually all subsequent studies in which tinnitus loudness was matched to external tones (40–43). To a person with normal hearing, a sound less than 10 dB above threshold would be only slightly perceptible and seemingly little cause for distress. It is thus a paradox that patients who report severe tinnitus usually match their tinnitus to these low-level sounds (41,44,45).

Numerous investigators have suggested that the inordinately low-level tinnitus LMs might be explained by a well-known phenomenon associated with hearing impairment—that is, the disproportionately rapid growth of loudness as sound intensity is increased (45–48). This loudness abnormality, which is referred to in the audiologic literature as “loudness recruitment,” is generally associated with reduced hearing sensitivity caused by cochlear pathology (49–52).

Goodwin and Johnson (46) evaluated the effect of loudness recruitment by matching tinnitus loudness at both the tinnitus frequency and at a normal-hearing frequency. For their subjects, LMs for tinnitus were obtained at higher sensation levels when measured at frequencies where hearing was normal (mean = 24 dB SL) than at the tinnitus frequency (mean = 7 dB SL). These results suggested that tinnitus loudness is significantly underestimated when measured at the tinnitus frequency. This same result has been replicated by others (53–55).

Tinnitus LMs have been used as an outcome measure to demonstrate the efficacy of drug treatment (e.g., [31]) and surgery (e.g., [32]). Johnson et al. (31) conducted a double-blind randomized clinical study to evaluate the effectiveness of Xanax (alprazolam) as a relief agent for tinnitus. A significant difference was observed between groups with respect to LMs—more subjects in the alprazolam group had reduced LMs, which was reported to support the effectiveness of using this drug for tinnitus. Cope, Baguley et al. (32) used LMs to document changes in loudness after resection of vestibular schwannoma. Although these studies assume that a reduction in the tinnitus LM reflects an actual reduction in tinnitus loudness, that assumption has not been verified.

More detailed information regarding tinnitus loudness matching is available elsewhere (56,57). The more salient points about tinnitus loudness matching include: 1) LMs are generally reliable within subjects (both within and between sessions), even if the matching tone does not sound like the tinnitus (55); 2) LMs have essentially no clinical value with respect to diagnosing tinnitus,

assessing its severity, determining the best form of intervention for an individual, or for measuring outcomes (56); and 3) LMs are often used by clinicians for counseling purposes (to “validate” the tinnitus perception and to point out that tinnitus is a “faint signal” even though it seems “loud”) (58).

Constrained loudness scaling is a method of psychophysical scaling developed by Ward and Baumann (59) to produce “more meaningful” measures of tinnitus loudness. A computer program trains the listener to perform the task by presenting a 1-second, 1000-Hz tone at 17 intensity levels. After each tone, the listener chooses a number from 1 to 100 to indicate the tone’s loudness. After each choice, the computer displays the “standard” number assigned to that level, which effectively “calibrates” the listener to the constrained loudness scale. After training, the listener makes a series of judgments, using the same scale, of the loudness of tinnitus, and of the loudness of a tone representing the dominant tinnitus pitch. The pitch-matched tone is presented at 17 intensity levels, interleaved with the 17 levels of a 1000-Hz tone, which maintains calibration on the standard response scale. Ward and Baumann’s initial data suggest that judgments of tinnitus loudness are valid using this technique, “yielding measurements that were substantially greater than the sensation level of sounds matched to tinnitus loudness” (p. 119). The constrained scaling method holds promise as providing valid measures of tinnitus loudness.

As a brief summary of tinnitus loudness measures: 1) tinnitus loudness cannot be objectively quantified; 2) loudness matching is helpful only for counseling purposes; 3) LMs and loudness ratings generally do not correlate; and 4) it is more useful clinically to obtain loudness ratings than LMs.

Measures of Tinnitus Pitch

Tinnitus pitch refers to its perceived frequency, or center frequency of a spectrum of nontonal tinnitus. The standard measure of tinnitus pitch is tinnitus pitch matching. For pitch matching, the frequency of a tone is varied and the patient selects the tone that best matches the pitch of the tinnitus. Most typically, patients match their tinnitus to a tone greater than 3 kHz (44).

It is generally agreed that an individual’s tinnitus pitch and audiometric configuration are typically related in some manner (2). The exact nature of the relationship, however, is unclear. Many investigators think that the pitch of tinnitus corresponds to the frequency region of the audiogram either at a point of maximum hearing loss or at the transition from normal to abnormal hearing (60–62). Meikle (63) summarized pitch-matching data from a sample of over 1,000 tinnitus patients to illustrate the considerable variability in the location of perceived tinnitus frequencies (cf. Figs. 14–3 and 14–5). Graham and Newby (40) earlier showed this same variability. About all that is certain is that tinnitus pitch matches (PMs) generally occur anywhere in frequency regions where there is hearing loss (56).

The pitch of tinnitus has been studied in relation to auditory conditions that are associated with tinnitus (56). Nodar and Graham (64) compared PMs between subjects with conductive hearing loss, sensorineural hearing loss, and Ménière’s disease. The subjects with conductive hearing loss had a median PM of 490 Hz (range 90–1450 Hz), which was clearly different from those with sensorineural hearing loss (median 3900 Hz, range 545–7500 Hz). These findings were consistent with those of Graham and Newby (40). All of the subjects with Ménière’s had PMs below 1000 Hz (median 320 Hz), which agreed with findings of numerous other investigators (65–67).

In the clinic, a single tinnitus PM is normally obtained and recorded as the patient’s perceived tinnitus frequency. When pitch matching is repeated, however (even within a session), the responses typically vary over 2 to 3 octaves (68–71). The poor reliability of tinnitus PMs could be because of a number of reasons, including: 1) patients may have difficulty with the PM task; 2) the tone used for matching may sound more like noise because of hearing loss (phenomenon of diplacusis, i.e., degraded pitch perception) (72); 3) tinnitus is likely experienced not as a pure tone but as a spectrum of sounds for most people (73,74).

In an effort to improve the reliability of tinnitus pitch matching, Penner and Bilger (75) used the forced-choice double staircase (FCDS) method to measure tinnitus pitch repeatedly. The FCDS procedure differs from tinnitus pitch matching in that patients classify a comparison stimulus relative to the tinnitus (higher or lower in pitch) rather than matching the stimulus to the tinnitus. Using this procedure, Penner and Bilger obtained standard deviations that were much smaller than those obtained in PM studies. The FCDS procedure as described in their study, however, would be impractical for clinical use because it is time-consuming and requires participants to be trained in the test paradigm. In an effort to adapt FCDS for clinical application, the procedure was modified to be conducted rapidly and without previous training (76). Results of testing, however, were unreliable, which was thought to be because of poor comprehension and fatigue on the part of the participants (the FCDS test came at the end of a full tinnitus psychoacoustic battery). Testing for tinnitus pitch with the FCDS procedure continues to hold promise as a reliable technique and further research is needed.

Tinnitus pitch matching continues to be widely used even though results are unreliable. It has been recommended that multiple PMs be used to more accurately identify the tinnitus frequency (77). These researchers developed a method of Bayesian sequential analysis to combine PMs until acceptable precision is achieved. In that pilot study, 30 tinnitus PMs were obtained using a computer-automated technique from each of 10 participants with chronic tinnitus. Results of testing revealed that 7 of the 10 participants provided tinnitus frequency estimates thought to be within one-quarter octave of their true value with 90% certainty. Between 4 and 20 PMs

were required to achieve acceptable results in these 7 patients. In addition, six of these seven subjects required only six PMs to achieve acceptably precise tinnitus estimates. Further development of this Bayesian approach to obtain tinnitus PMs may enable a level of precision that has not previously been achieved.

To briefly summarize tinnitus pitch matching: 1) PMs tend to occur on the edge or in the middle of the hearing loss region; 2) any report of “tinnitus pitch” should be considered an estimate that is within a 2 to 3 octave range; 3) tinnitus is most often a “spectrum” of sound, and should be measured for spectral content.

Tinnitus Spectrum Match

Traditional pitch matching procedures, as described above, use pure tones as test stimuli. These procedures identify a single-center frequency to represent the pitch of tinnitus. Another approach is to rate a number of frequencies for their resemblance or “likeness” to the tinnitus sound (73,74). Such testing produces a “tinnitus spectrum” that typically encompasses the sloping region of the audiogram, consistent with the spread of repeated tonal pitch reported by others (68–71). It has been reported that tinnitus spectra peak at significantly higher likeness ratings in the hearing loss region for subjects reporting a “tonal” (pure-tone) tinnitus versus those reporting tinnitus that sounds more like noise (74). Studies have also reported good test–retest reliability when likeness ratings were correlated: 1) between sessions for each tested frequency (74); and 2) across frequencies within a subset of individual subjects (78). We conducted a study that included testing to determine if a person’s tinnitus sounds more like a tone or more like a band of noise (79). Most subjects in that study provided reliable responses, both within and between sessions, with respect to the type of stimulus (tonal, narrow band, or wide band) selected as the best match to their tinnitus. These studies support the contention that the tinnitus percept often (perhaps most often) has “spectral content,” which may help to explain why patients tend to be inconsistent when performing PMs with pure tones; that is, if the tinnitus consists of a spectrum of frequencies, repeated PMs with pure tones would be expected to span the range of frequencies. For this reason, psychoacoustic measures of tinnitus should include some form of spectral matching to provide for more accurate characterization of the tinnitus percept.

Minimum Masking Level

In the clinic, minimum masking level (MML) refers to the minimum level of broadband noise (BBN) required to render a patient’s tinnitus inaudible. Whereas measures of tinnitus loudness and pitch (and spectrum) attempt to quantify the tinnitus percept, MML measures the effect of sound on the percept of tinnitus. Hazell et al. (34) suggested referring to this as the minimum *suppression* level because “suppression” more accurately reflects the effect of sound suppressing the tinnitus neural signal. Nevertheless, minimum masking level is the terminology that remains in common use.

It would do an injustice to discuss tinnitus masking without referring to Feldmann’s “masking curves.” In 1971, Feldmann reported a study originally intended to reveal that masking of tinnitus would result in conventional masking patterns, i.e., when masker intensity is plotted as a function of frequency, the intensity of the masking tone is lowest when its frequency is close to that of the tone being masked (80). If so, Feldmann thought this could provide an alternative method for determining the pitch of tinnitus. He obtained masking patterns from 200 subjects, and observed five different types of masking patterns—none of which resembled conventional masking curves.

Feldmann also found that 11% of his subjects were “unmaskable,” i.e., none of his masking stimuli were able to fully mask these subjects’ tinnitus. He further observed that any weak sound could effectively mask the tinnitus for 32% of his subjects. Surprisingly, for some of his subjects the presentation of sound in one ear masked the tinnitus in the contralateral ear. He concluded that neural activity responsible for tinnitus differed from neural activity resulting from stimulation with external sound. His was the first of many subsequent studies all showing that perception of tinnitus does not follow the same psychoacoustic principles that apply to the perception of external sounds (56).

Data from the Oregon Health and Science University Tinnitus Clinic show that tinnitus can be completely masked by BBN presented at 6 dB SL for 42% of patients, and within 12 dB SL for 70% of patients (81). These data suggest that tinnitus is easily masked for most patients. Other reports, however, suggest that many patients with bothersome tinnitus perceive it most of the time, even in loud environments (82).

MML testing is commonly done by audiologists as part of a tinnitus psychoacoustic assessment. The objective of MML testing is to determine the lowest level of BBN that renders tinnitus inaudible (complete elimination of tinnitus percept). Tinnitus can also be partially masked, meaning stimulation with sound causes spectral changes in the tinnitus percept and/or reduced perception of tinnitus loudness.

Measures of MML were purported to be prognostic with respect to which patients would be most likely to benefit from treatment with tinnitus masking (38,83). Basically, patients whose tinnitus was most easily masked were most likely to benefit, whereas those least able to be masked were least likely to benefit. The testing is done by presenting a BBN to one or both ears. Thus, monaural or binaural MMLs can be obtained. For tinnitus sound therapy in general, MMLs provide an indication of the effectiveness of environmental sound for making a patient’s tinnitus “less noticeable.”

As a brief summary, the following points can be made about MML testing: 1) masking effects are unpredictable. Patients range from “easily masked” to “unmaskable.” It is unknown how the tinnitus neural signal of these patients might be different. 2) MML testing is done in some clinics, and the test results have been used to assist

in determining the effectiveness of masking as treatment. 3) Changes in MML have been cited as evidence of improvement as a result of intervention for tinnitus (50). Use of MML data in this manner can lead to misleading reports of benefit derived from the intervention. Because it is possible to have a high MML without substantial annoyance to the patient, MMLs simply are not accurate indicators of emotional distress or functional effects of tinnitus. Consequently, changes in the MML cannot be relied upon as a valid indicator of intervention outcomes. 4) MML testing has little if any value for the tinnitus assessment. Clinicians should refrain from using results of MML testing to assess outcomes, and instead use tools designed to directly assess functional effects of tinnitus.

Residual Inhibition

Residual inhibition (RI) refers to the temporary suppression (reduction in loudness/intensity) or elimination of tinnitus that often occurs after auditory stimulation (84,85). Feldmann (80) is well known for his systematic study of the masking of tinnitus, which revealed characteristic types of tinnitus “masking curves.” In the same report, he noted how “the tinnitus remains silent for a certain period of time after cessation of the inhibitory stimulus” (p. 142). Feldmann’s “inhibitory stimuli” were 500 msec bursts of 1/3-octave noise bands centered at 1 kHz with varying intervals of silence between the noise bursts. Feldman found that his subjects typically could not detect their tinnitus when intervals of silence between the noise bursts were as long as a second, and sometimes longer. Feldmann concluded “If this mechanism could be trained or activated there might be a way to cure patients of their distressing tinnitus” (p. 144). The effect was named *residual inhibition* by Vernon and Schleuning (86) in recognition of Feldmann’s work. RI has been shown to occur with most patients who receive the appropriate type of auditory stimulation (44,87–89). It is therefore surprising that the phenomenon has received little study with regard to specific parameters of acoustic stimuli that induce RI.

The clinical test for RI is to present BBN (2–12 kHz) binaurally to patients at 10 dB above their MML (84,85). The noise is delivered for 60 seconds, and then terminated abruptly at which time patients are asked to describe any perceived changes to their tinnitus. If the tinnitus is suppressed, then they are asked to comment regarding changes to the tinnitus percept during tinnitus recovery to its normal level. These comments and the duration of the effect to full recovery are recorded. The occurrence of RI is a consistent observation when effective tinnitus masking is performed. Using the standard clinical test for RI, some degree of RI is reported to occur for 80 to 90% of patients with tinnitus (38). RI has been reported to last less than 2 minutes in 60% of patients and less than 4 minutes in 80% (81). For many patients it is the first time they have experienced a silencing of their tinnitus since its inception, and they can be overcome with emotion. It should be noted that a patient’s tinnitus *can* get louder after RI testing.

Despite the clear potential to experience RI during treatment with ear-level devices, very few patients actually benefit from this effect (38,84,85,90). Vernon and Meikle (84) stated “... if one could shape the masking band of noise to be optimum for any given patient, one might be able to extend greatly the duration of residual inhibition and conceivably even transform it into a permanent or semipermanent condition” (p. 51). It seems a worthwhile effort, therefore, to develop standard clinical procedures, based on experimental data, for identifying stimuli that maximize RI. The stimulus that maximizes RI in an individual could then be presented using a protocol that maintains a prolonged state of RI with minimal stimulation.

In summary, some important facts have emerged from the limited RI studies. First, the phenomenon of RI is repeatable within a given individual; that is, RI can be produced repeatedly as often as the stimulus is presented for some minimum duration (91–93). Second, after RI the tinnitus returns to its previous level of intensity within a period that is typically quite brief. For the majority of patients, tinnitus returns to its normal level within about 1 minute. A small percentage of individuals, however, will experience RI for 5 to 30 minutes or even longer—occasionally the tinnitus is gone for several days. Despite the obvious importance of the phenomenon, it is at present unknown to what degree RI can be enhanced in the average tinnitus patient. There is clearly the potential for RI to be used as a clinical therapeutic technique, provided the optimal stimuli can be identified and then presented in wearable devices.

Because RI is such a repeatable phenomenon, occurs in 80 to 90% of tinnitus patients, and actually reduces the sensation of tinnitus (unlike other tinnitus therapies), it is important that studies be designed and conducted to evaluate the temporal, spectral, and intensity parameters of acoustic stimuli that influence its behavior. The proper study of RI will require implementing a systematic algorithm using a variety of stimulus parameters to determine if and how the effect might be enhanced to achieve long-term suppression. If these parameters could be identified, and the specified sound stimuli could be presented directly to the ears of patients, the inducement of RI could become a viable method of treatment.

Need for Normative Data

Many tinnitus researchers routinely obtain measures of tinnitus perception (31–34). However, normative data do not exist to facilitate interpretation of the measures—other than the Oregon Tinnitus Data Archive (OTDA), which is available as an online data registry. The OTDA contains data from 1,630 patients observed at the Oregon Health and Science University Tinnitus Clinic between 1983 and 1994, and serves as a valuable resource for tinnitus clinicians and researchers. The OTDA psychoacoustic data, however, have significant limitations: 1) data are combined across all age groups—age-specific norms are not available; 2) test equipment varied; 3) different clinicians used different test procedures; 4)

pitch matching was limited to the use of pure tones (noise-band matching was not done); 5) PMs were only obtained one time for each patient, thus not addressing the known poor reliability of PMs.

Normative tinnitus psychoacoustic data are important to identify relationships between measures of tinnitus perception and other factors that could help to: 1) elucidate underlying mechanisms of tinnitus; 2) facilitate a more definitive assessment of the association between these measures and measures of tinnitus impact; and 3) predict specific types of therapy that might be most beneficial to an individual patient.

Objective measures do not exist to assess compensation claims involving tinnitus. Dobie (54) stated “The problem of assessing injury and damages without the assistance of objective standards is particularly acute where claims of tinnitus are included. The lack of reliability of measurement of loudness and pitch by matching techniques and the prevalence of tinnitus in the population compounds the problem” (p. 243). The establishment of normative standards would enable tinnitus psychoacoustic measures to be evaluated with respect to a set of norms.

CONCLUSIONS

To date, there has been no known method for reducing the perception of tinnitus, which would normally be experienced as a reduction in tinnitus loudness. The problem with using psychoacoustic measures to assess outcomes of treatment for tinnitus is thus twofold: 1) the measures have not been shown to correlate with changes in functional effects of tinnitus; and 2) methods do not exist to suppress or eliminate (i.e., cure) tinnitus. For these reasons, outcomes assessment in tinnitus research relies mainly on participants’ subjective ratings of functional effects of (or reactions to) tinnitus. Research is needed to develop methods to: 1) objectively quantify tinnitus loudness; 2) obtain reliable measures of tinnitus dominant pitch; 3) describe the spectrum of tinnitus; 4) use RI as a clinical technique. Further, on the basis of studies of neural plasticity, it has been hypothesized that tinnitus perceptual attributes can be modified. Research is needed to use auditory training procedures to suppress tinnitus in the long term.

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